Causal Check on Othello-GPT: Separating Downstream World-Model Computation from Direct Readout Bias

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1. The Question and Why It Matters for World Models Now

A central claim in world-model interpretability is that self-supervised sequence models learn linear internal representations of world state that causally drive next-step predictions. In **Othello-GPT**, Nanda, Lee & Wattenberg (2023) showed that probing residual activations in player-relative coordinates ({`Empty, Yours, Mine`}) yields $> 99\%$ linear probe accuracy by Layer 4, and that adding probe vectors across layers steers the model into predicting legal moves for a counterfactual board B'. However, two questions remain open whenever linear activation steering alters a transformer's output:

1. **Mediated computation vs. direct readout bias:** Because a transformer's residual stream forms an identity skip connection to the final unembedding ($\mathbf{x}_L = \mathbf{x}_\ell + \sum_{k=\ell}^{L-1} f_k(\mathbf{x}_k)$), injecting a steering vector $\boldsymbol{\delta}_\ell$ across layers $\ell \in \{2, \ldots, 7\}$ also adds $\sum_\ell \boldsymbol{\delta}_\ell$ directly into the final readout W_U. Does steering succeed because intermediate Attention/MLP layers recompute Othello capture rules over the edited board state, or because $\boldsymbol{\delta}$ acts as a direct linear logit bias at W_U?
2. **Single-layer decodability vs. causal control:** If a linear probe decodes the board state with 99.4% accuracy at Layer 5, does intervening at Layer 5 alone flip the model's move predictions, or do downstream layers overwrite single-layer state edits?

Why it matters now: Linear probe R^2 and activation steering are widely used to certify that video, robotics, and market models have learned actionable world states. If a representation is 99% decodable at layer ℓ yet causally overwritten unless edited across a specific circuit span—or if steering metrics are confounded by direct skip-connection bias—standard probe diagnostics will overstate whether a world model can be planned with under intervention.

2. Method: Models, Data, Metrics, and Controls

- **Model & Data:** We audit the pretrained 8-layer, 8-head, $d_{\text{model}} = 512$ `Othello-GPT` (`synthetic_model.pth`, `LNPre`) and the 8 relative linear probes `resid_{0..7}_linear.pth` $\in \mathbb{R}^{512\times8\times8\times3}$ from `ajyl/mech_int_othelloGPT`. Probing is evaluated on 3,584 mid-game states ($t \in [5, 50]$) across 80 held-out synthetic games. Causal interventions are evaluated on the complete official benchmark (`intervention_benchmark.pkl`, $N = 1{,}001$ partial games) from Li et al. (2023) and Nanda et al. (2023), where flipping a single occupied square $s = (r, c)$ (`Yours` $\leftrightarrow$ `Mine`) changes the ground-truth legal move set from $M(B)$ to $M(B')$.
- **Metrics (1,000-sample bootstrap 95% CIs across games):** (1) **Counterfactual Move Error** (Nanda et al. Table 2 metric, lower is better): for $K = |M(B')|$ and top-K model predictions $\hat{M}_K$, $\text{Error}(B') = |\hat{M}_K \setminus M(B')| + |M(B') \setminus \hat{M}_K|$. (2) **Newly-Legal Probability Gain (ΔP_{new}):** change in probability mass assigned to newly legal moves $M_{\text{new}} = M(B') \setminus M(B)$, $\Delta P_{\text{new}} = \sum_{m \in M_{\text{new}}} [P(m \mid \text{intervened}) - P(m \mid \text{clean})]$.
- **Controls that rule out the boring explanations:**
  1. **Direct-Logit Bypass Control vs. Skip-Subtracted (`Mediated-Only`):** To test whether $\boldsymbol{\delta}$ merely biases W_U via the skip connection, `Direct-Logit Bypass` leaves hidden layers $\ell \in \{0, \ldots, 7\}$ clean and adds $\sum_{\ell=2}^{7} \boldsymbol{\delta}_\ell$ *only* to the final residual stream before `ln_final` $+ W_U$. Conversely, `Mediated-Only` injects $+\boldsymbol{\delta}_\ell$ at layers $\ell \in \{2, \ldots, 7\}$ so downstream layers compute over the edited state, then *subtracts* $-\sum_{\ell=2}^{7} \boldsymbol{\delta}_\ell$ before `ln_final` $+ W_U$ so the direct skip path is zero.
  2. **Untrained-Encoder Probe Control:** Fit a ridge linear probe on a randomly initialized 8-layer GPT of identical architecture and variance (200 training games) and steer `Othello-GPT` along the resulting probe directions.
  3. **Wrong-Square Spatial Control ($s' \neq s$):** Steer using the trained probe vector of a non-intervened interior square $s' \neq s$ at the same norm $\alpha = 2.3$.

3. First Experiment and Results

Experimental Arm ($N = 1{,}001$ boards, $\alpha = 2.3$)	**Error on B' ↓ (95% CI)**	**ΔP_{new} ↑ (95% CI)**	**Finding / Role**
0. Null Baseline ($\alpha = 0$)	2.723 [2.645, 2.799]	+0.000 [+0.000, +0.000]	Exact match to Nanda et al. Table 2 (2.723)
1. Full Multi-Layer Steering (Nanda et al. `intervene.py`)	0.102 [0.072, 0.136]	+0.072 [+0.066, +0.077]	Exact match to Nanda et al. Table 2 (0.10)
2. **Mediated-Only (Skip-Subtracted before W_U)**	**0.158** [0.122, 0.198]	**+0.037** [+0.033, +0.041]	**97.9% of error drop survives removing skip path**
3. **Direct-Logit Bypass Control (Only before W_U)**	1.808 [1.730, 1.884]	**+0.000** [+0.000, +0.000]	**Zero probability gain on newly legal moves**
4. Untrained-Encoder Probe Control	2.661 [2.581, 2.743]	+0.000 [+0.000, +0.000]	Random-init probe directions have no causal effect
5. Wrong-Square Spatial Control ($s' \neq s$)	3.435 [3.337, 3.530]	+0.001 [+0.001, +0.002]	Steering an unrelated square degrades $M(B')$
6. Code Audit A: Layer-Matched `probes[`ℓ`]` @ `attn_out`	0.150 [0.116, 0.188]	+0.071 [+0.065, +0.076]	Replaces hardcoded `probes[5]` in `intervene.py`
7. Code Audit B: Layer-Matched `probes[`ℓ`]` @ `resid_post`	0.076 [0.054, 0.102]	+0.075 [+0.069, +0.081]	Aligns intervention hook with probe training site
8. **Code Audit C: Contrastive Flip @** `resid_post`	**0.012** [0.004, 0.026]	**+0.079** [+0.073, +0.085]	8.5× **error reduction (12 errors / 1,001 boards)**

Table 1: Causal decomposition and codebase audit on the official 1,001-board Othello-GPT benchmark.

- **Result 1 — Downstream computation, not direct logit bias, drives counterfactual steering (Fig. 1A, Table 1):** Removing the direct skip-connection vector before `ln_final` $+ W_U$ (`Mediated-Only`) reduces counterfactual error from 2.723 to 0.158 (97.9% of the full reduction). Conversely, injecting the same accumulated vector only before W_U (`Direct-Logit Bypass`) produces $\Delta P_{\text{new}} = +0.000$: direct unembedding dot products slightly suppress invalidated moves (1.808 vs. 2.723) but cannot identify newly legal captures without intermediate Attention/MLP layers.
- **Result 2 — High single-layer decodability ≠ single-layer causal control (Fig. 1B):** Trained 3-way probe accuracy reaches 99.0% at `L4` and 99.5% at `L6` (vs. 63.4% → 62.6% on the untrained encoder, which decays monotonically as random layers scramble token history). Yet editing a single layer alone never pushes error below 2.188 (`L5`), because downstream layers re-read un-edited board features from earlier residual streams and overwrite the single-layer edit. Cumulative span edits [ℓ..7] show that causal consumption peaks at **Span `[L3..L7]` (0.072 error)** and collapses if started at `L5` (1.345) or `L6` (2.268): by the layer where probe accuracy peaks (`L6`), the capture-detection circuits in Layers 3–4 have already fired on the un-intervened board.

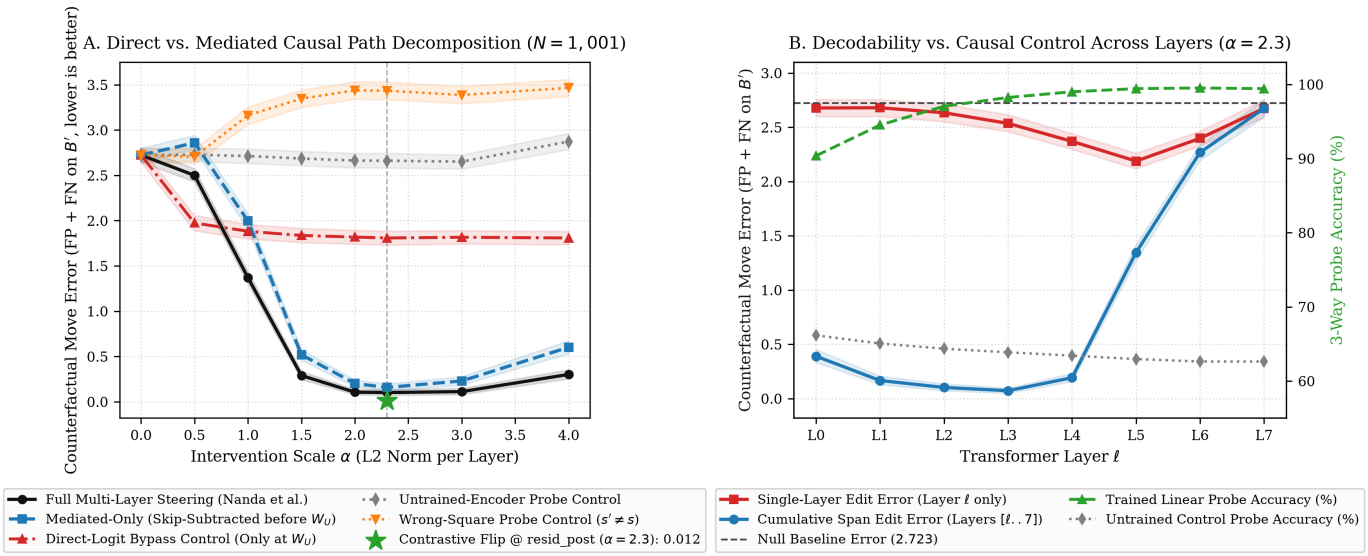


Figure 1: Left (A): Counterfactual move error across intervention scale α ($N = 1{,}001$, shaded 95% bootstrap CIs). Subtracting the direct skip connection (`Mediated-Only`, blue) preserves 97.9% of the error reduction, while adding the vector only before W_U (`Direct-Logit Bypass`, red) fails to discover newly legal moves. Fixing the hook site and using a contrastive flip vector ($\star$, green) cuts error from 0.102 to 0.012. **Right (B):** Layer-wise 3-way probe accuracy (green/gray, right axis) vs. single-layer edit error (red) and cumulative span [ℓ..7] error (blue).

- **Result 3 — Auditing `ajyl/mech_int_othelloGPT` cuts steering error by** $8.5\times$ **(**$0.102 \to 0.012$**):** Inspecting `mech_int/intervene.py` (lines 140–162) revealed three discrepancies between Nanda et al.'s paper and code: (i) the script constructs per-layer hooks `hook_fns[0..7]` but hardcodes `hook_fns[5]` (Layer 5's probe) into all layers 2..7; (ii) it hooks `hook_attn_out` (pre-MLP) using probes trained on `resid_post` (post-MLP); and (iii) it adds only the target class vector $\hat{\mathbf{w}}_{\text{target}}$ without subtracting the old state $\hat{\mathbf{w}}_{\text{source}}$. Hooking `resid_post` with layer-matched probes drops error to 0.076, and steering along the **contrastive flip direction** $\hat{\mathbf{d}}_s^\ell \propto \mathbf{w}_{s,\text{target}}^\ell - \mathbf{w}_{s,\text{source}}^\ell$ reduces error from **0.102** to **0.012** [**0.004**, **0.026**]—only 12 errors across all 1,001 boards.

4. What Result Would Change My Conclusion, and What I Would Run Next Week

- **Pre-Registered Falsification Criteria:** (1) Our conclusion that steering is driven by mediated computation rather than readout bias would be falsified if freezing hidden Attention/MLP outputs to clean activations while applying $\boldsymbol{\delta}$ before W_U recovered $> 50\%$ of the error reduction or yielded $\Delta P_{\text{new}} > +0.02$ (it yields 33.6% error reduction and $+0.000$ ΔP_{new}). (2) Our conclusion that single-layer edits fail due to downstream self-repair from earlier residual streams would be falsified if projecting out the old tile subspace at Layer 3 *and* blocking Layers 4–6 attention heads from attending to the historical move token where s was played still failed to reduce single-layer edit error below 0.50.
- **Next Week's Plan:** (1) **Path-patching Attention vs. MLP self-repair:** Decompose the gap between `Single-Layer L3` (2.535) and `Span [L3..L7]` (0.072) by patching individual MLP blocks vs. attention heads in Layers 4–6 to separate MLP local-pattern heuristics (Lin et al. 2024) from attention re-reading. (2) **Endgame `MoveFirst` regime ($t > 30$):** Stratify the 0.012 contrastive intervention by game turn t and compare discriminative probe weights against difference-in-means vectors ($\boldsymbol{\mu}_{\text{Mine}} - \boldsymbol{\mu}_{\text{Yours}}$) in late-game states where Nanda et al. found moves are computed before full board state.

5. What I Want to Work On With Physical Reasoning

I want to work on **Evaluating World Models Under Intervention**—specifically claiming and building `physicalreasoning/forward` **Issue #5 (*"Design an intervention test for world models"*)**, and combining it with the probing controls from `causal-jepa`. In my research at the **CausalML Lab** (modeling multi-agent failures as soft interventions on Structural Causal Models) and on **hierarchical traffic scene synthesis**, models with nearly identical observational validation loss routinely diverge under counterfactual interventions. Concretely, I want to build a lightweight, deterministic 2D rigid-body/traffic counterfactual benchmark for `forward` where two small world models trained to matched observational validation loss are reset to held-out states, subjected to unshown impulse/mass interventions, and scored under a pre-registered gate on the **counterfactual consistency gap** alongside direct-vs-mediated and untrained-encoder controls.

Code & Reproducibility: Standalone script (`run_experiment.py`), Othello simulator (`othello_rules.py`), and raw bootstrap outputs (`results.json`) reproduce all numbers and figures in $\sim$ 3 minutes on CPU or $<$ 20s on GPU. **AI Tools Disclosure:** AI coding assistants (Antigravity / Gemini) were used to scaffold the pure-PyTorch `LNPre` forward pass, KV-cached batching, and Matplotlib layout; all experimental design decisions, control formulations, codebase audits of `ajyl/mech_int_othelloGPT`, and interpretations were verified directly against the executed benchmark outputs.